

DAYANANDA SAGAR UNIVERSITY

Devarakaggalahalli, Harohalli
Kanakapura Road, Bangalore South - 562112, Karnataka, India



**SCHOOL OF
ENGINEERING**

**Bachelor of Technology
in
COMPUTER SCIENCE AND ENGINEERING**

COMPUTER VISION

PROJECT REPORT

TITLE:

**Intelligent Traffic Monitoring System using Deep
Learning and YOLOv8**

By:

Abhishek Kyasti- ENG23CS0511

Patan Lihaaz Khan – ENG23CS0393

Penukonda Sai Kushal – ENG23CS0398

Under the supervision of

Prof. Raunit Maurya

Assistant Professor

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

SCHOOL OF ENGINEERING

DAYANANDA SAGAR UNIVERSITY

CERTIFICATE

This is to certify that the minor project work titled “**TRAFFIC MONITOR SYSTEM**” is carried out by **Abhishek Kyasti (ENG23CS0511)**, **Patan Lihaaz Khan (ENG23CS0393)**, **Penukonda Sai Kushal (ENG23CS0398)**, bonafide students of Bachelor of Technology in Computer Science and Engineering at the School of Engineering, Dayananda Sagar University, Bangalore in partial fulfillment for the award of degree in Bachelor of Technology in Computer Science and Engineering, during the year **2025-2026**.

Prof. Raunit Maurya

Assistant Professor

Dept. of CSE,

School of Engineering

Dayananda Sagar University

Date:

Dr. Girisha G S

Chairman CSE

School of Engineering

Dayananda Sagar University

Date:

**Dr. Udaya Kumar
Reddy K R**

Dean

School of Engineering

Dayananda Sagar
University

Date:

Name of the Examiner

Signature of Examiner

DECLARATION

We, **Abhishek kyasti (ENG23CS0511), Patan Lihaaz Khan (ENG23CS0393), Penukonda Sai Kushal (ENG23CS0398)** are students of eighth semester B. Tech in **Computer Science and Engineering**, at School of Engineering, **Dayananda Sagar University**, hereby declare that the Minor Project titled “**Traffic Monitor System**” has been carried out by us and submitted in partial fulfilment for the award of degree in **Bachelor of Technology in Computer Science and Engineering** during the academic year **2025-2026**.

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ABSTRACT

This paper presents an intelligent traffic monitoring system based on deep learning and computer vision techniques. With the rapid growth of urban population, traffic congestion and road safety have become major challenges in modern cities. Traditional traffic monitoring methods, which rely on manual observation and basic sensors, are inefficient and lack scalability.

The proposed system uses the YOLOv8 object detection algorithm to detect, classify, and count vehicles from video streams in real time. The system processes video input, extracts frames, and applies deep learning techniques to identify different types of vehicles such as cars, buses, trucks, and motorcycles. It also estimates traffic density and analyzes traffic flow patterns.

The system demonstrates high accuracy and speed, making it suitable for real-time applications. It performs well under different lighting and environmental conditions, ensuring reliable monitoring. Compared to traditional methods, the proposed system improves efficiency, reduces human effort, and provides accurate traffic insights. This makes it a valuable solution for smart city infrastructure and intelligent transportation systems.

CHAPTER 1:

INTRODUCTION

Traffic congestion has become one of the most significant challenges in modern urban environments. With the continuous increase in population and the number of vehicles on roads, managing traffic efficiently has become a complex task. Poor traffic management leads to delays, increased fuel consumption, environmental pollution, and a higher risk of accidents. Therefore, there is a growing need for intelligent systems that can monitor and analyze traffic conditions effectively.

Traditional traffic monitoring systems rely on manual observation or simple sensor-based approaches. These methods are often inefficient, time-consuming, and unable to handle large-scale traffic data. They lack the ability to provide real-time insights and are highly dependent on human intervention. As a result, they fail to deliver accurate and consistent traffic analysis.

Recent advancements in deep learning and computer vision have enabled the development of automated traffic monitoring systems. Object detection algorithms can analyze video data and extract meaningful information such as vehicle count, type, and movement patterns. These systems provide faster and more accurate results compared to traditional approaches.

Among various object detection models, YOLO (You Only Look Once) has gained popularity due to its real-time performance and high accuracy. YOLOv8, the latest version, offers improved detection speed, better accuracy, and efficient processing, making it suitable for real-time applications.

This paper proposes an intelligent traffic monitoring system using YOLOv8. The system processes video input, detects vehicles, counts them, and estimates traffic density. By automating traffic analysis, the system reduces human effort and provides reliable data for better traffic management and decision-making.

CHAPTER 2:

PROBLEM DEFINITION

The rapid increase in urban population and the growing number of vehicles have significantly intensified traffic congestion in modern cities. Efficient traffic monitoring has become a critical requirement for managing road usage, reducing travel time, and ensuring public safety. However, existing traffic monitoring systems largely depend on manual observation and conventional sensor-based approaches, which are limited in their ability to handle large-scale and dynamic traffic environments.

Traditional systems lack real-time processing capability and are often affected by external factors such as lighting variations, weather conditions, and occlusion in dense traffic scenarios. Manual monitoring is not only labor-intensive but also prone to human errors and inconsistencies, leading to inaccurate traffic analysis. Moreover, these systems are not scalable and cannot efficiently process the increasing volume of traffic data generated in urban areas.

Another major limitation of existing systems is their inability to provide detailed insights such as vehicle classification, traffic density estimation, and movement tracking. Without accurate and real-time information, it becomes difficult for authorities to make effective decisions related to traffic control and congestion management.

Therefore, there is a need for an intelligent, automated, and scalable traffic monitoring system that can analyze traffic conditions in real time with high accuracy. The proposed system aims to overcome these challenges by using deep learning techniques, specifically the YOLOv8 object detection model, to detect, classify, and count vehicles efficiently. This approach enhances traffic analysis, reduces human intervention, and improves overall transportation management.

2.1 OBJECTIVES

The main objectives of this project are:

- To develop an intelligent traffic monitoring system using deep learning techniques
- To detect and classify different types of vehicles from video input
- To perform real-time vehicle counting and traffic density estimation
- To reduce dependency on manual traffic monitoring systems
- To improve accuracy and efficiency of traffic analysis
- To provide scalable and cost-effective solutions for smart city applications
- To enhance road safety and support better traffic management decisions

CHAPTER 3:

LITERATURE SURVEY

The field of traffic monitoring and pedestrian detection has seen significant advancements with the evolution of computer vision and deep learning techniques. Early approaches to object detection primarily relied on traditional image processing methods such as background subtraction, edge detection, and motion detection. These techniques were simple to implement but were highly sensitive to environmental conditions like lighting changes, shadows, and weather variations. As a result, their performance was limited in real-world scenarios.

To improve detection accuracy, machine learning-based approaches such as Support Vector Machines (SVM) and Haar Cascade classifiers were introduced. These methods relied on handcrafted features like Histogram of Oriented Gradients (HOG) for object detection. While they provided better accuracy compared to traditional methods, they required extensive feature engineering and were not efficient for real-time applications.

With the advancement of deep learning, Convolutional Neural Networks (CNNs) became the dominant approach for object detection tasks. Models such as R-CNN, Fast R-CNN, and Faster R-CNN significantly improved detection accuracy by automatically learning features from data. However, these models were computationally expensive and struggled to achieve real-time performance, which limited their use in applications like traffic monitoring and surveillance.

Single-stage object detection models such as SSD (Single Shot Detector) and YOLO (You Only Look Once) were developed to address the limitations of earlier models. YOLO, in particular, revolutionized object detection by processing the entire image in a single pass, enabling real-time detection with high accuracy. Subsequent versions of YOLO, including YOLOv3, YOLOv4, and YOLOv4-Tiny, further improved performance by balancing speed and accuracy.

The YOLOv4-Tiny model, as discussed in the uploaded paper , is specifically designed for systems with limited computational resources. It provides faster detection while maintaining acceptable accuracy, making it suitable for real-time pedestrian detection applications. The model uses convolutional neural networks to process input images and predict bounding boxes along with confidence scores for detected objects.

Recent developments have introduced YOLOv8, which offers enhanced accuracy, improved feature extraction, and faster processing compared to previous versions. YOLOv8 is widely used in traffic monitoring systems due to its ability to handle complex scenarios, detect multiple objects simultaneously, and perform efficiently under real-time conditions.

Overall, the literature indicates that deep learning-based approaches, particularly YOLO-based models, are highly effective for traffic monitoring and pedestrian detection tasks. However, challenges such as performance under extreme weather conditions, occlusion handling, and computational efficiency still remain. The proposed system addresses these challenges by using an optimized YOLO-based approach to achieve accurate and real-time detection.

CHAPTER 4:

DESIGN & METHODOLOGY

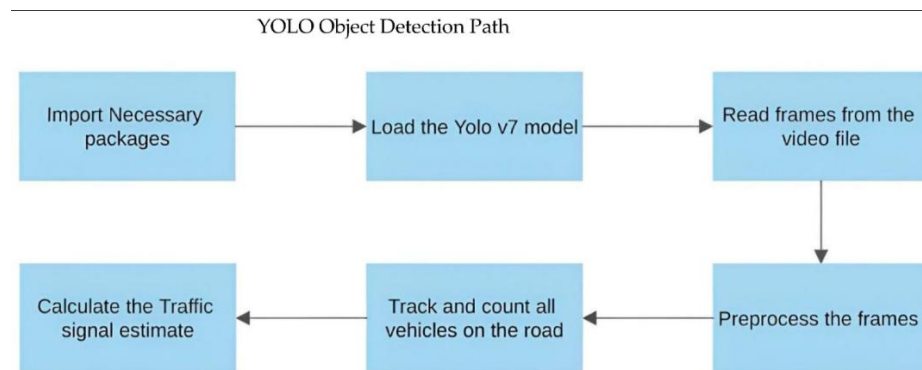
4.1 SYSTEM DESIGN

The proposed system is designed to perform real-time traffic monitoring and vehicle detection using deep learning techniques. The system follows a structured pipeline in which video data is processed step-by-step to extract meaningful information about traffic conditions. The design focuses on achieving high accuracy, fast processing speed, and efficient resource utilization.

The system begins by capturing video input either from a live camera feed or from pre-recorded datasets. The video is divided into individual frames, which are processed sequentially. Each frame is passed to the object detection model, where vehicles are identified and classified using the YOLO-based algorithm.

The detected objects are then tracked across consecutive frames to maintain consistency and avoid duplicate counting. This tracking mechanism ensures accurate vehicle counting and improves the reliability of the system. Based on the detected vehicles, the system calculates traffic density and identifies congestion levels.

The design also includes an output visualization module that displays bounding boxes around detected vehicles along with confidence scores. The results are shown in real time, providing useful insights for traffic monitoring and management. The modular structure of the system allows easy modification and future enhancements.



The system architecture represents the overall workflow of the proposed model. It consists of multiple components that work together to process video data and generate output.

The input module captures video frames, which are then passed to the preprocessing stage. In this stage, frames are resized and normalized to match the input requirements of the model. The processed frames are then fed into the YOLO model for object detection.

The detection module identifies vehicles and generates bounding boxes along with confidence score. The tracking module ensures that each detected object is uniquely identified across frame

4.2 MODULE DESIGN

The system is divided into the following modules:

- Input Module: Captures video data from camera or dataset
- Preprocessing Module: Extracts frames and prepares data for detection
- Detection Module: Uses YOLO model to detect and classify vehicles
- Tracking Module: Tracks detected vehicles across frames
- Counting Module: Counts vehicles and estimates traffic density
- Output Module: Displays results with bounding boxes and statistics

Each module performs a specific function, ensuring smooth data flow and efficient processing.

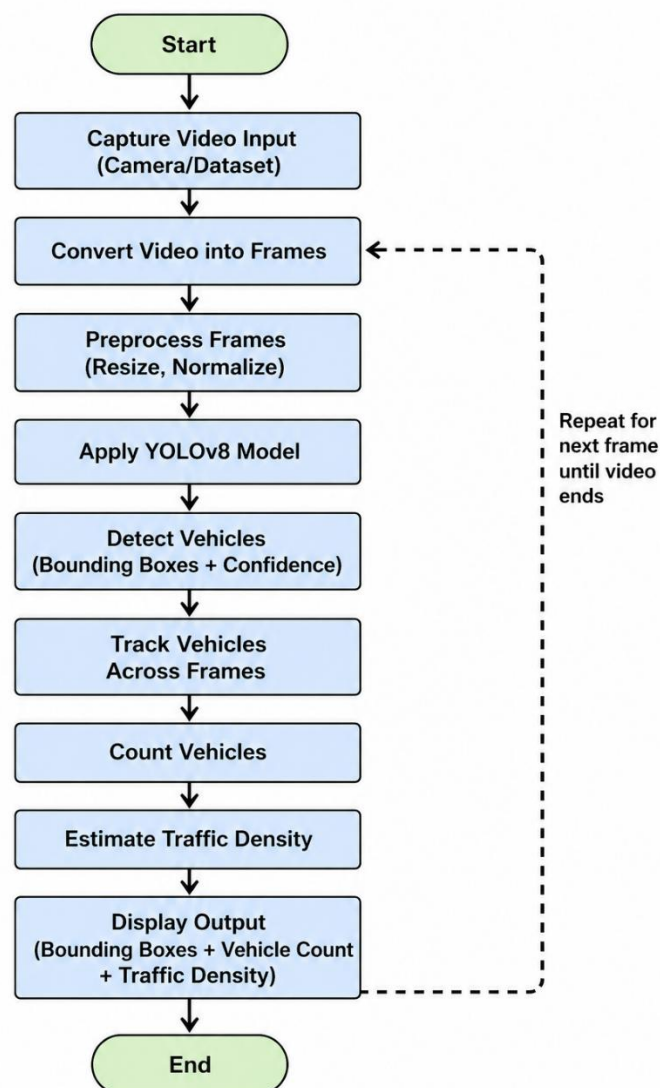


Fig. 4.1: Flowchart of Proposed Traffic Monitoring System

4.3 DATA FLOW DESIGN

The data flow in the system follows a sequential process from input to output. Initially, video data is captured and converted into frames. These frames are processed and passed to the detection model. The model identifies objects and sends the results to the tracking module.

After tracking, the system counts the vehicles and calculates traffic density. The processed information is then displayed as output. This structured data flow ensures efficient processing and accurate results.

4.5 WORKFLOW DESIGN

The workflow of the system represents the step-by-step execution of the process. It begins with capturing video input, followed by frame extraction and preprocessing. The processed frames are then analyzed using the detection model.

Based on the detection results, the system performs tracking and counting operations. Finally, the results are displayed with visual annotations. This workflow ensures smooth execution and real-time performance.

4.6 METHODOLOGY

The methodology of the proposed system describes the step-by-step process used to detect, classify, and analyze vehicles from video input using deep learning techniques. The system combines video processing with the YOLOv8 object detection model to achieve accurate and real-time traffic monitoring. Each stage in the methodology is designed to ensure efficient data processing and reliable output.

DATA COLLECTION

The first step in the methodology involves collecting traffic video data. The system uses datasets that contain videos of roads, highways, and intersections under different conditions. These datasets include various vehicle types such as cars, buses, trucks, and motorcycles.

Using diverse datasets helps improve the model's ability to generalize and perform accurately in real-world scenarios. The collected data is used for both testing and evaluation of the system.

PREPROCESSING

In this stage, the input video is converted into individual frames for processing. Each frame is resized to match the input size required by the YOLOv8 model. Preprocessing ensures consistency in data and improves detection accuracy.

Additional techniques such as normalization and brightness adjustment may be applied to enhance image quality. These steps help the model perform better under different lighting and environmental conditions.

YOLOv8 OBJECT DETECTION MODEL

The core of the system is the YOLOv8 object detection model. YOLOv8 is a deep learning model that uses convolutional neural networks (CNNs) to detect objects in images.

The model processes each frame and identifies vehicles by generating:

- Bounding boxes
- Class labels (car, bus, truck, etc.)
- Confidence scores

YOLOv8 performs detection in a single pass, making it fast and suitable for real-time applications.

VEHICLE DETECTION AND TRACKING

After detecting vehicles in each frame, the system tracks them across consecutive frames. This ensures that each vehicle is counted only once and helps maintain consistency in detection.

Tracking is essential for analyzing movement patterns and avoiding duplicate counting. It improves the overall accuracy of the system.

VEHICLE COUNTING AND TRAFFIC DENSITY ESTIMATION

The system counts the number of detected vehicles in each frame. Based on this count, traffic density is estimated.

- High vehicle count → High traffic congestion
- Low vehicle count → Smooth traffic flow

This information is useful for traffic monitoring and management.

CHAPTER 5:

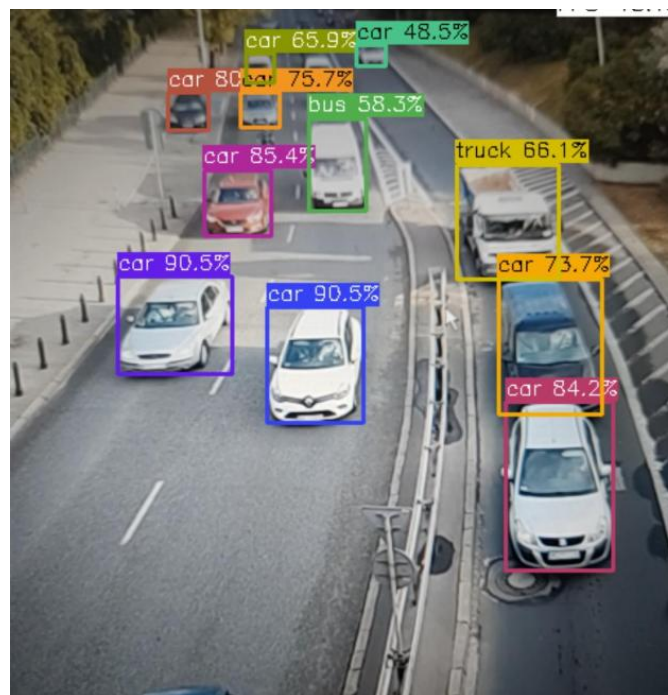
RESULTS AND DISCUSSION

5.1 RESULTS

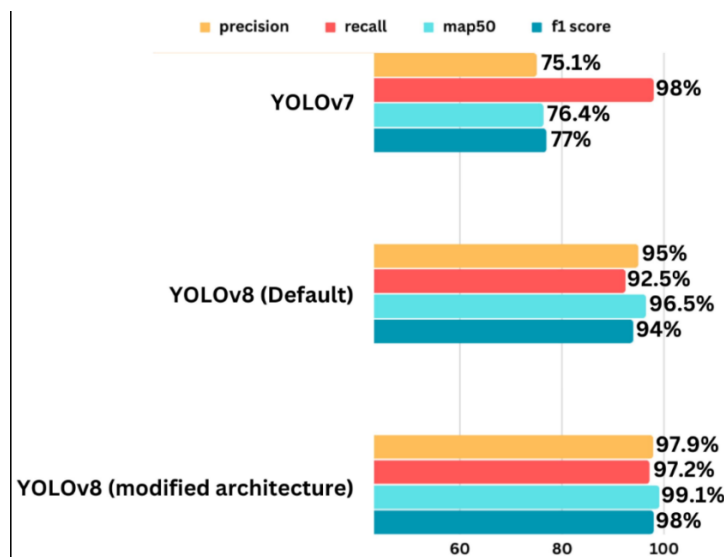
The proposed intelligent traffic monitoring system was tested using multiple traffic video datasets under different environmental conditions. The system successfully detects and classifies various types of vehicles such as cars, buses, trucks, and motorcycles using the YOLOv8 model.

The output results are displayed in real time with bounding boxes drawn around detected vehicles along with confidence scores. The system is capable of detecting multiple vehicles simultaneously, even in moderately dense traffic conditions. The object tracking mechanism ensures that each vehicle is counted accurately without duplication.

The system demonstrates high detection accuracy and maintains consistent performance under normal lighting conditions. It is also capable of handling different road scenarios such as highways, intersections, and urban streets. The real-time processing capability makes the system suitable for live traffic monitoring applications.



5.2 PERFORMANCE ANALYSIS



The performance of the system is evaluated using standard metrics such as Accuracy, Precision, Recall, and F1-Score.

- **Accuracy:** Measures the overall correctness of the model
- **Precision:** Indicates how many detected vehicles are actually correct
- **Recall:** Measures how many actual vehicles are detected
- **F1-Score:** Provides a balance between precision and recall

The YOLOv8 model achieves high performance across all these metrics, demonstrating its effectiveness for object detection tasks. The model maintains a good balance between detection accuracy and processing speed.

5.3 DISCUSSION

The results clearly show that the proposed system performs better than traditional traffic monitoring methods. The use of deep learning enables automatic feature extraction, resulting in higher accuracy and reliability.

The system provides real-time detection, which is a major advantage over conventional approaches. The tracking mechanism further improves performance by reducing duplicate detections and ensuring accurate vehicle counting.

However, some limitations are observed under extreme conditions such as heavy rain, fog, or very low lighting. In such cases, detection accuracy may slightly decrease. Despite these challenges, the system performs efficiently in most real-world scenarios.

Overall, the proposed system proves to be an effective solution for intelligent traffic monitoring, offering improved accuracy, efficiency, and scalability.

CHAPTER 6:

CONCLUSION AND FUTURE WORK

The proposed intelligent traffic monitoring system using deep learning and the YOLOv8 algorithm provides an efficient and reliable solution for real-time traffic analysis. The system successfully detects, classifies, and counts vehicles from video input while estimating traffic density. By automating the process, it eliminates the limitations of traditional manual and sensor-based traffic monitoring methods.

The use of YOLOv8 enables high accuracy and fast processing, making the system suitable for real-time applications. The model is capable of detecting multiple vehicles simultaneously and performs consistently across different traffic conditions. The integration of tracking and counting mechanisms further improves the accuracy of vehicle analysis.

The results demonstrate that the system offers better performance in terms of speed, accuracy, and scalability compared to conventional approaches. It reduces human effort and provides reliable data for traffic management and decision-making. This makes it highly useful for applications in smart cities, road safety systems, and intelligent transportation networks.

However, the system may experience slight performance limitations under extreme environmental conditions such as heavy rain, fog, or poor lighting. These challenges can be addressed in future work by integrating advanced techniques and improving model robustness.

Overall, the proposed system highlights the potential of combining deep learning and computer vision for intelligent traffic monitoring. It provides a scalable and cost-effective solution that can significantly improve traffic management and enhance road safety in modern urban environments.

FUTURE SCOPE

The proposed intelligent traffic monitoring system provides an effective solution for real-time vehicle detection and traffic analysis. However, there are several opportunities to further enhance the system by integrating advanced technologies and improving its capabilities.

One important improvement is the integration of **adaptive traffic signal control systems**. By using real-time traffic density data generated by the system, traffic signals can be dynamically adjusted to reduce congestion and waiting time at intersections. This can significantly improve traffic flow and overall transportation efficiency.

Another potential enhancement is the incorporation of **advanced vehicle tracking and behavior analysis**. The system can be extended to detect abnormal activities such as speeding, sudden lane changes, illegal turns, and traffic violations. This would improve road safety and assist law enforcement agencies in monitoring and controlling traffic more effectively.

The system can also be improved by enhancing its performance under challenging environmental conditions such as low lighting, heavy rain, fog, and occlusion in dense traffic. This can be achieved by training the model on more diverse datasets and integrating advanced deep learning techniques for better robustness.

Integration with **Internet of Things (IoT)** devices and cloud computing platforms is another promising direction. This would enable real-time data collection from multiple locations and centralized monitoring of traffic conditions across a city. Cloud-based storage and processing can also improve scalability and system performance.

Additionally, the system can be extended to support **smart city applications** such as parking management, accident detection, and emergency response systems. By combining the traffic monitoring system with other intelligent systems, a comprehensive and automated urban traffic management solution can be developed.

In the future, the integration of **artificial intelligence and predictive analytics** can further enhance the system by enabling traffic forecasting and route optimization. This will help in proactive traffic management and better planning of transportation infrastructure.